

Nikhil Prabhu

(925) 819-4040 | nikhilprabhu06@gmail.com | [linkedin.com/in/nikprabhu1](https://www.linkedin.com/in/nikprabhu1) | github.com/Trolleroof | nikhilprabhu.tech

EDUCATION

University of California, San Diego (UCSD)
B.S. in Computer Science | Cumulative GPA: 3.67

San Diego, CA
Expected Graduation: June 2028

SKILLS

Languages: Python, C++, TypeScript, JavaScript

Deep Learning: PyTorch, Gymnasium, vision-language pipelines, behavioral cloning, Gemini / VLM integration

RL & Policies: PPO, MAPPO, reward design, sim-to-real, online rollout/eval loops, MCP agent tooling

Robotics + Infra: ROS 2, SLAM, visual odometry, MuJoCo manipulation sims, FastAPI, Docker, WebSockets

EXPERIENCE

Undergraduate Research Assistant — AI for Circuit Invention, UCSD

San Diego, CA

Research Assistant (with Prof. Prasad Gudem & Samvid AI)

March 2026 – Present

- Designed an LLM-driven generation pipeline with a closed-loop Cadence evaluator, iterating on model behavior against simulation feedback — the same generate, evaluate, and refine cadence used to train robotic policies.
- Exploring reinforcement learning strategies for design-space search; building tooling to benchmark model outputs at scale against simulation baselines.

Lead Researcher — Brain-Computer Interface (Stanford Lee-Messer Lab)

Stanford, CA

Stanford University School of Medicine

Oct 2023 – Dec 2024

- Built and benchmarked ML models on a P300 EEG paradigm, turning noisy multimodal sensor streams into reliable real-time control commands.
- Owned the full research loop — data collection, model iteration, and deployment — piloting with 12 users on a shipped React Native assistive platform.

Lead Researcher — Foot Drop Ankle-Foot Orthosis (Georgia Tech mmWave Lab)

Atlanta, GA

IEEE ICHCI 2024 Presenter & Published Author

May 2024 – Nov 2024

- Built an EMG-controlled actuation system ($R = 0.968$ correlation, 23% gait precision gain); first-author paper published and presented at IEEE ICHCI 2024.

ROBOTICS & RL PROJECTS

DroneOS

1st Overall, Bow Capital $\times$ DS3 $\times$ SIC Hackathon (UCSD)

2026

- Trained multi-agent swarm policies (MAPPO + behavioral cloning) for GPS-denied navigation and target identification, validating learned behavior in simulation before deployment.
- Built the perception stack: YOLOX + Norfair detection/tracking fused with SLAM, visual odometry, and 3D Gaussian Splatting for friend-or-foe labeling on edge.

Apollo Labs & Battle Angel — RL Policy Training

1st Overall, OpenClaw Hackathon & Google Gemini 3.5 Hackathon

2026

- Built a FastAPI PPO orchestrator with RunReports, rollout GIFs, versioned artifacts, and MCP tools for autonomous policy optimization.
- Trained Unitree G1 PPO agents across eight MuJoCo disaster environments with language-driven rollouts in unseen scenes — directly adjacent to home-robot autonomy policy design.

SODIUM — Embodied Home Robot

1st Overall, DiamondHacks 2026

April 2026

- Built a voice-first in-home care robot on ROS 2 with wake-word detection, NL task routing, and a real-time caregiver dashboard — full embedded-to-cloud autonomy loop.

Motion — Robot Training Data Pipeline

Stanford $\times$ DeepMind Hackathon

2026

- Built a crowdsourced manipulation marketplace with Gemma 4 + Kimi K2 VLM verification labeling clips for fleet-scale policy training (LeRobot / Open X-Embodiment export).

PUBLICATIONS

- Real-Time EMG Control for Assistive Devices** — IEEE ICHCI 2024.
- Co-Heritability of Antibiotic Resistance in TB Evolution** — Bioinformatics & Biomedical Technology, 2025.